

The Ephemeris

March 2023

Volume 34 Number 01- The Official Publication of the San Jose Astronomical Association



March 2023 - June 2023 Events

Schedule subject to change

Please check our website for up to date information.

Board & General Meetings

Saturday 3/4, 4/8, 5/6, 6/3

Board Meetings: 6 - 7:30pm

General Meetings: 8:00 - 9:30pm

Solar Sunday (2:00 - 4:00pm)

Sunday 3/5, 4/2, 5/7, 6/4

In-Town Star Party (7:00-9:00pm)

Friday 3/24, 4/21, 5/12, 6/16

RCDO Starry Nights Star Party (7:00 - 9:00pm)

Saturday 3/11, 4/15, 5/13, 6/10

Imaging SIG (online)

Tuesday 3/21, 4/18, 5/16, 6/20

Astro Imaging Clinics (at Little Uvas OSP)

Saturday 3/18, 4/22, 5/20, 6/17

Binocular Star Gazing (at Coyote Valley OSP)

Saturday 6/10

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

INSIDE THIS ISSUE

Annual Board and Officer Elections	2
In Memoriam Paul Mancuso	3
School Star Parties	4
Article	6
Astro Photos	7
Articles	11
Kid Spot	13
Board of Directors	14
Club Programs	15
Membership Form	16

California Nebula
Photo by SJAA Member: Jian Peng
December 27, 2022
Location: Palo Alto, CA

© Jian Peng Astrophotography

SJAA Board and Officer Elections

This year at the February 4th 2023 board meeting, SJAA announced the results of the board elections. The meeting was held online. Directors Vini Carter, Ken Miura, Swami Nigam, Peter Melhus and Sukhada Palav were reelected in their positions. During the March 4th 2023 board meeting the board elected the new SJAA Officers. Aroon Nair was elected as President, Chris Gottbrath was elected as Vice-President, Rob Jaworski was reelected as Treasurer and Swami Nigam was elected as Secretary. Heartiest congratulations to all elected!



Introducing Chris Gottbrath - Vice-President

Hello all! I'm Chris Gottbrath, your new Vice-President. Astronomy has been a life long passion for me. I can't say for sure where it started -- probable candidates include: Star Trek, NASA Space Shuttle missions, that beat up copy of Carl Sagan's Cosmos on my shelf, a family friend who lent me a 6 or 8" dob for a high school science project, or simply seeing the Milky Way span from one horizon to the over midwest corn-fields. All I can say for sure was that I picked my goal early. Throughout my school years I told everyone in a confident tone that I would be a professor of astronomy someday soon and I hiked quite a way down that path. Life had other things in store though and I've subsequently spent more than 20 years in the tech industry -- often talking with scientists -- sometimes even astronomers -- but as a vendor, industry collaborator, and technology specialist rather than a colleague.

A great thing about astronomy though is that it makes a fantastic hobby with a low entry bar and a ramp that goes on and on. I've got a motley collection of telescopes and cameras, and am eagerly learning how to pull structure and beauty out of what I can capture with them. I'm a supporter and organizer of the University of Arizona hands-on Astronomy Camp Program (www.astronomycamp.org) which has been inspiring middle and high school age scientists for 35 years and have recently had a lot of fun with robotic imaging from great sites in both hemispheres with the iTelescope network (www.itelescope.net).

I've been a member of the San Jose Astronomy Association for 6 years since being introduced to it by Francesco but have never taken advantage of the opportunity to really dig into everything the association offers. Aroon's call for volunteers caught my eye this winter and I decided the time was right to step up and help out. I am excited for the chance to get to know everyone here better, indulge my curiosity about the universe, and apply myself to fostering the unique community that is the SJAA.



Introducing Jonathan Lawton - QuickSTART Leader

After several "hobby-killer" experiences with department store telescopes growing up, Jonathan's interest in astronomy was rekindled as an adult when a friend gave him a short binocular astronomy book. His first sight of the Andromeda Galaxy through binoculars was the beginning of his journey in the hobby which now includes: deep sky object imaging, visual observing, public outreach, and more. Jonathan is excited to share his passion for the night sky with the public and to help newcomers to the hobby get started confidently. He lives in the hills between San Jose and Morgan Hill with his wife and two young children.

In Memoriam Paul Mancuso

Credit: Rob Jaworski



In December 2022, the SJAA lost a long-time member, volunteer and friend, Paul Mancuso. Paul started his SJAA journey way back in the 1970s. He was very involved with the club, over the years serving as board member and in various officer roles. He kept his involvement going over the last 20 years as a stalwart volunteer at the School Star Parties as well as maintaining a regular presence at the In Town Star Parties at Houge Park.

Paul's ashes have been interred at Cedar Lawn Cemetery, in the Silayan Garden, Mausoleum II. SJAA members attended the internment service and were a comfort to his wife and daughter. His family invites all SJAA members and friends to visit there and pay respects at any time. They are located at 48800 Warm Springs Blvd, Fremont.

The following are remembrances of Paul from SJAA members.

The school star party program was an important outreach. Among the group of volunteer astronomers, some showed up sporadically, others were regulars. Paul Mancuso was one of the regulars. There were a few school star parties where just three astronomers showed up. Paul was one of those stalwarts. As proof of that, in 2016, Paul was awarded

the SJAA's highest honor, the Gregory Award.

<https://www.sjaa.net/about/gregory-award/>

The last school star party of the 2017-2018 season was April 24, 2018, and of course, Paul was there. It was at Thomas Russell Middle in Milpitas, just a few blocks from Paul's house as it turned out. We were almost clouded out, the sky was thin overcast. The Moon barely shown through, only a few of the brightest stars were visible some of the time. Still the kids were interested despite not being able to see much. As usual we all made the best of it, answering questions and talking about astronomy. In hindsight now, it seems that weather was a portent.

This was the last school star party of the season and possibly the last of the school star party program altogether since Jim Van Nuland was retiring as our coordinator after some 30 years. After the kids were gone and we were all packed up, we paused for a moment looking at each other. The realization started setting in that this might be the end of the school star party program. We shook hands and talked about what a really great run it had been. After a few minutes of some reflection on a very worthy program well done, we said good-bye and went home. Of course the public star parties would continue at least, maybe the school star parties too if we get a new coordinator. So we never thought that was really "good-bye."

Not surprisingly the 2018-2019 season wasn't much, we presented at only a few schools, and then the pandemic hit. It turns out we really were saying good-bye after all. At least we got to do that. Many don't get that chance. I'm sure all the many kids who got to look through Paul's scope will remember the good impression he made on them.

—Gary Mitchell

I used to run into Paul at Houge Park often, where he would set up at the north end of the sidewalk (Telescope Row). For years, I only knew him by voice, and every time, he would seem to have a different scope set up from the last time. Paul was great with the public as he shared views at his eyepiece, and he was also adept with the kids at the schools. That's not to mention that Paul was an all-around good guy with a great sense of humor. He'll be missed.

—Rob Jaworski

School Star Parties

Credit: Marianne Damon

SJAA provides outreach to local schools via our School Star Party (SSP) program. For years the SSP was organized by Jim Van Nuland, then by Cat Cvengros Slack, and recently by Jon Anderson. Schools request a date via our website page and then the school liaison and Jon work out the logistics. After a hiatus during Covid, schools are now able to hold these events once again. The “season” generally runs from October through March, when the evenings are dark early enough for young children to be out, and then home before bedtime. Jon did several by himself to get the program rolling again. The program has lately been gifted a couple of scope donations from Rich Gregor-- thanks, Rich!

The first announced SSP of the year was on October 6 at P.A. Walsh Academy in Morgan Hill. There was a large turnout with some of the teachers and parents bringing their own scopes. The sky cooperated with a clear beautiful evening displaying Jupiter, Saturn and the Moon which elicited cries of oohs and awes from attendees. Paul Kohlmeier and Jon each brought their 10-inch Meade LX-200s and Marianne had her BT-70 to round out the viewing. We were thanked by everyone for providing a great night of astronomy and the school is already planning their next SSP.

The second one was at Discovery Charter Falcon Elementary in San Jose. When we arrived, the field was full of camping tents and the children were having a talent show. Families were having dinner, some brought from home, and some purchased at a BBQ catering truck. The three of us (Jon, Paul, and Marianne) set up away from the activities on the basketball court. When it was dark enough, we turned on our red lantern which was the signal for people to start coming over. We had a steady stream of enthusiastic children and parents to view Jupiter, Saturn, and various colored stars and to point out constellations and answer a wide range of questions. Many children, after viewing, ran off to bring their parents over to view! The number of people observing was evened out by the S'mores activity, with people going back and forth, but luckily, we did not have a problem with sticky fingers. We packed up at 10 pm, just as families were settling into their tents and a gorgeously orange waning gibbous Moon was rising.

Weather cancelled the next SSP on November 1st, but at least the children had a program during the day where they got to talk with an astronaut!

The third star party, Tacos and Telescopes! was at Peterson Middle School on November 17. They were very happy to have us back after not being able to hold the event since the winter of 2019. The afternoon was cloudy but, since they were having dinner and astronomy related crafts and student speakers, three of us decided to attend and support the event. SJAA member Sathya Venkataraman and his son, who attends Peterson, set up their Orion Mak 150 mm Skyview Pro. Luckily, the sky opened up and we were able to show the students and their parents some wonderful views.

2023 brought our deluge of January rain. One school cancelled their evening, but on Saturday January 21 Hacienda Elementary hosted an amazing event! This school is an environmental science magnet and has won designation as a California Distinguished School. They have a one-acre outdoor classroom site featuring California native plants. From their website: “Special field trips to local sites expand students’ environmental education. And our annual Star Party is a stellar way for students’ eyes and minds to reach far out across space and time.” Yes, they are a “regular customer” of ours (although this was the first one since Covid). This year the families heard talks from NASA speakers, had photo opportunities, and planted tomato seeds that had flown in space! There were eight NASA/Ames scientists and engineers on hand, giving demonstrations on infrared imaging and what is a vacuum (the marshmallows got squished!). Former astronaut and Ames employee Steve Smith, who grew up in the neighborhood, talked about what it was like to live, work and walk in space. Outdoors, they were treated to views from two evScopes and six optical scopes; participating were our own Jon Anderson, Jonathan Lawton and Joe Trdnich, and members from PAS (Peninsula Astronomical Society), TAC (The Astronomy Connection) and Foothill College. All-in-all, over six hundred K-5 students and their family members had a stellar evening.

We conducted a school Star Party on Thursday Feb 2 at Campbell School of Innovation. We were hopeful for breaks in the clouds, but there were very few. The Moon was visible for about 20 minutes, then made random appearances through small breaks in the cloud cover. We were amazed with how many students and their parents took it positively, even lining up just to look through our scopes at the clouds. This was a fairly large event and our very own Jordan Makower provided talks with handouts in a classroom for rotating groups of students and families. Marianne demonstrated how to use binoculars, handed out magazines and answered questions. A parent brought their scope. Several families brought their table-top scopes and had set up on the picnic tables outside the auditorium. There were students learning about rocketry in a classroom also. The kids & parents had lots of questions and seemed very interested in not only the scopes but astronomy in general. A “Gang of Dads” made plans to join us for the first time at a Starry Nights at RCDO.

Mattos Elementary really, really, really wanted a school star party on Wednesday, February 22. There were clouds, but enough patches of sky coming and going to be able to show multiple objects. Everyone had been warned about the iffy situation, but enthusiasm was high and the turnout was huge. The organizer of the event had pizza and hot tea for us, and plenty of glowsticks for the kids. The combination of telescope, evScope and binoscope gave people a nice sample of astronomy gear. Every family received an astronomy magazine. Just as we packed up our gear and started our vehicles, the first big plops of rain came down. We felt very fortunate to have had a good night with a timely ending!

There are a few more school star parties scheduled before the end of the season, and we're hoping the skies will be clear enough for us to wow a few hundred more families.

I'll use this opportunity to put out a call for astronomy magazines. I give them out at our school star parties, one per family. I have a stack of my own from the pandemic hiatus but will gladly take any off your hands. I look through them to toss out advert cards, and I peel off the address labels or black out the name & address. Typical are S&T, Astronomy, Star Date, Planetary Report, The Reflector...save them for me! Let me know when you have a stack ready to unload, and we'll work out a rendezvous. And, if you have experience using your equipment for public events, please consider helping out with our school star parties! Contact Jon and he'll put you on the notification list. Thank You!



Above: Jon Anderson, Joe Trdinich, Marianne Damon and Mattos Elementary School Organizer and her daughter.

Photo taken on: February 22, 2023 at Mattos Elementary School in Fremont, CA

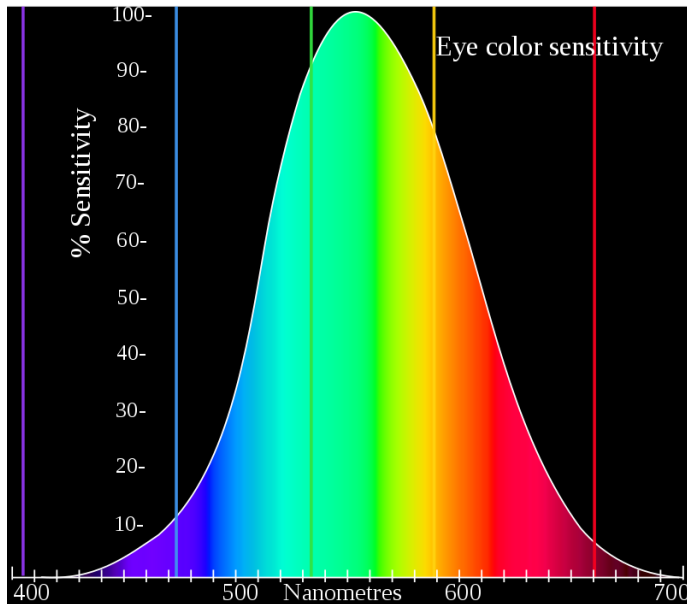
Credit: Marianne Damon

The Lives of the Stars

Credit: Jordan Makower

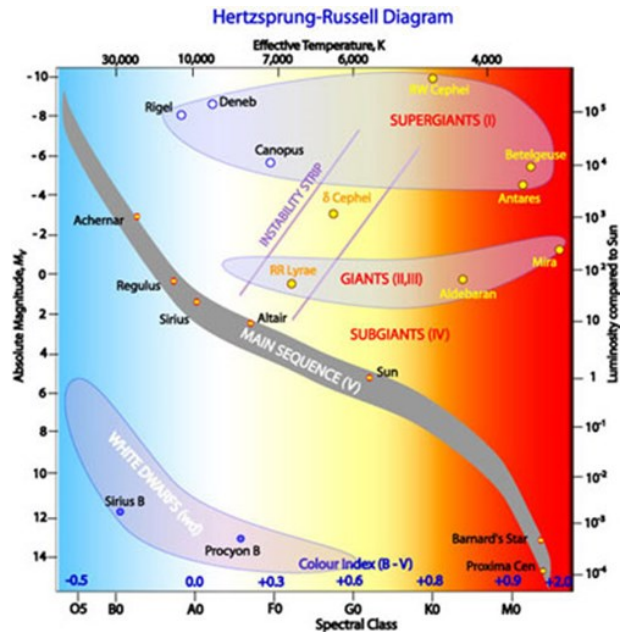
When a person casually gazes at the stars, they may be only slightly aware that some of them have faint colors. White stars seem to be the most plentiful, but there seem to be a few pinkish-red ones, too. Our eyes may not be discriminating enough to easily pick out yellow stars, like our Sun, and yet they are most sensitive to the yellow and blue parts of the energy spectrum. We can't, unaided, perceive the ultraviolet, infrared, radio, or other parts of the great electromagnetic radiation emitted by those stars, so it is fortunate that we have devices to take up where our senses leave off.

Stars are placed on a chart (the Hertzsprung-Russell Diagram) based on their true brightness (Absolute Magnitude; negative numbers indicate the brighter ones) and surface temperature (redder colors are the cooler ones). There are a great variety of stars represented there, and their placement at a particular phase of its life is also an indicator of its past and future.



Caption: Eye Color Sensitivity

Credit: Wikipedia



Hertzsprung—Russel Diagram

Credit: Cosmos

Scientists believe that the first stars were made of 'neutral' hydrogen, and as gravitational forces pulled more matter together, four hydrogens 'fused' forming helium, and releasing energy. The other elements were formed in subsequent fusions. A star stably spends most of its life on the Main Sequence area of hydrogen fusion. Ones with great amounts of matter go through this phase relatively quickly, perhaps in hundreds of thousands of years. Those with less matter may take millions or billions of years. Some stars expand and contract as they generate energy by fusion in their cores. The diagram shows representatives of stars of different masses in their life stages of generating energy. See if you can find some of these as you do your starwatching.



Date: January 23, 2023

Location: Home

Imaging Telescope: William Optics RedCat 71

Imaging Camera: ZWO ASI533MC Pro

Mount: ZWO AM5

Software: Adobe Photoshop · Pleiades Astrophoto PixInsight



Dates: February 9, 2023

Location: Home Base, Union City, CA

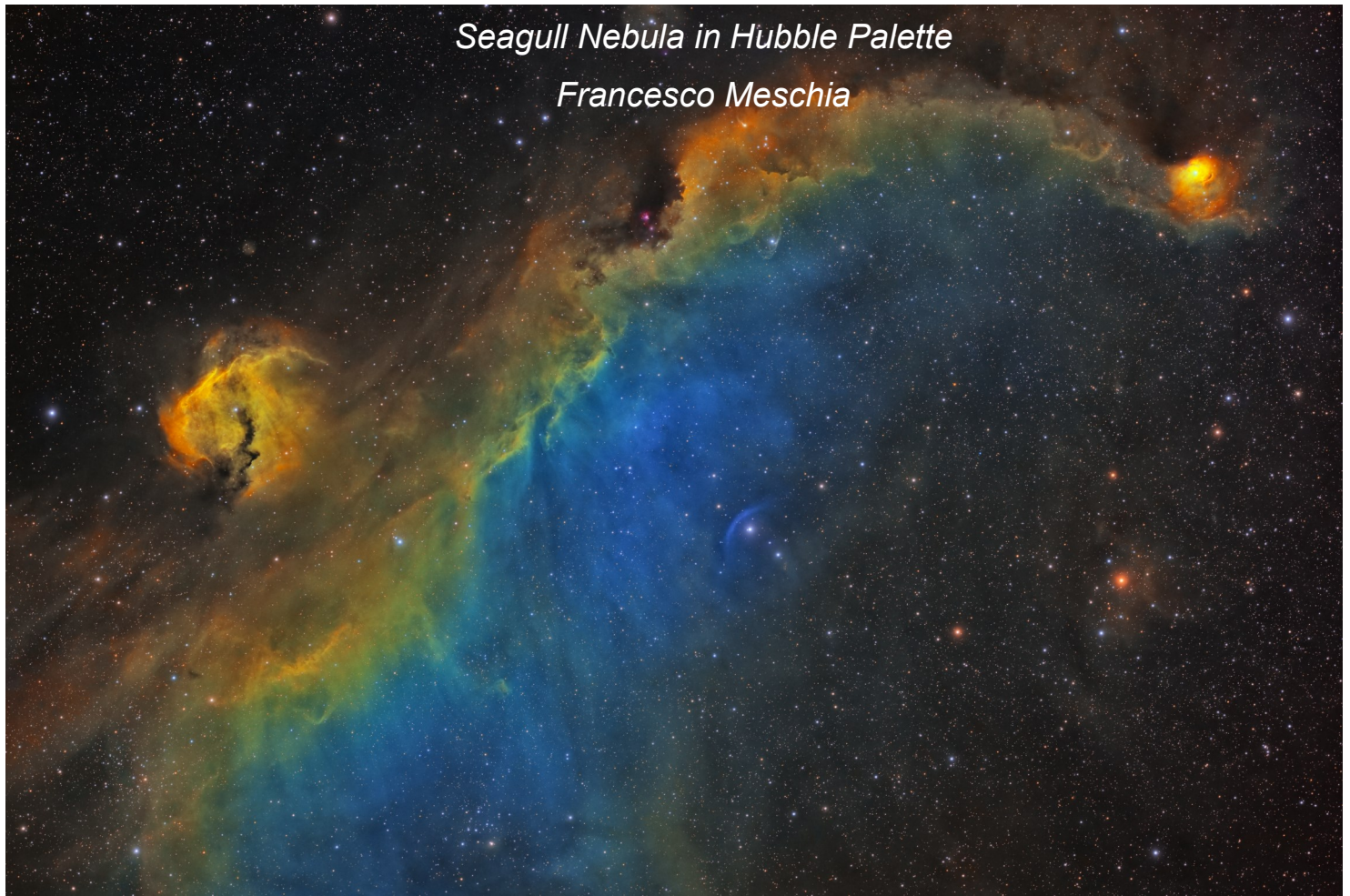
Telescope: TPO 12" Truss Tube f/8 Ritchey Chretien

Camera: ZWO ASI294MM Pro

Mount: Orion HDX 110 EQG

Software: PixInsight, Adobe Photoshop

SJAA Member Astrophoto Gallery



Date: Jan. 26, 2023, Jan. 27, 2023, Feb. 7, 2023, Feb. 8, 2023, Feb. 9, 2023

Location: Home, Mountain View, CA, United States; Russian Ridge OSP, Palo Alto, CA

Telescope: Astro-Tech AT65EDQ

Camera: ZWO ASI294MM Pro

Mount: iOptron iEQ45 Pro

Software: Cranfield / Payne Generalized Hyperbolic Stretching (GHS) Module/Script , Pleiades Astrophoto Pix-Insight, Russell Croman Astrophotography BlurXTerminator, Russell Croman Astrophotography NoiseXTerminator, Russell Croman Astrophotography StarXTerminator, Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NINA)

SJAA Member Astrophoto Gallery



Date: December 14-17, 25, 2022 and January 18-20, 22-25 2023

Location: Ladera, California

Telescope: GSO 10" f/8 Ritchey-Chretien Truss Tube

Camera: ZWO ASI1600MM Pro

Mount: Astro-Physics 1100GTO

Software: Adobe Lightroom Classic, Adobe Photoshop, KDEdu KStars, Pleiades Astrophoto PixInsight

Dynamic NASA-Built Weather Sensors Enlisted to Track Tropical Cyclones

Credit NASA



COWVR, at center, and TEMPEST, not shown, were installed aboard the International Space Station in late 2021 and since then have provided valuable weather data to forecasters tracking tropical cyclones. The two instruments are part of the U.S. Space Force STP-H8 demonstration mission.

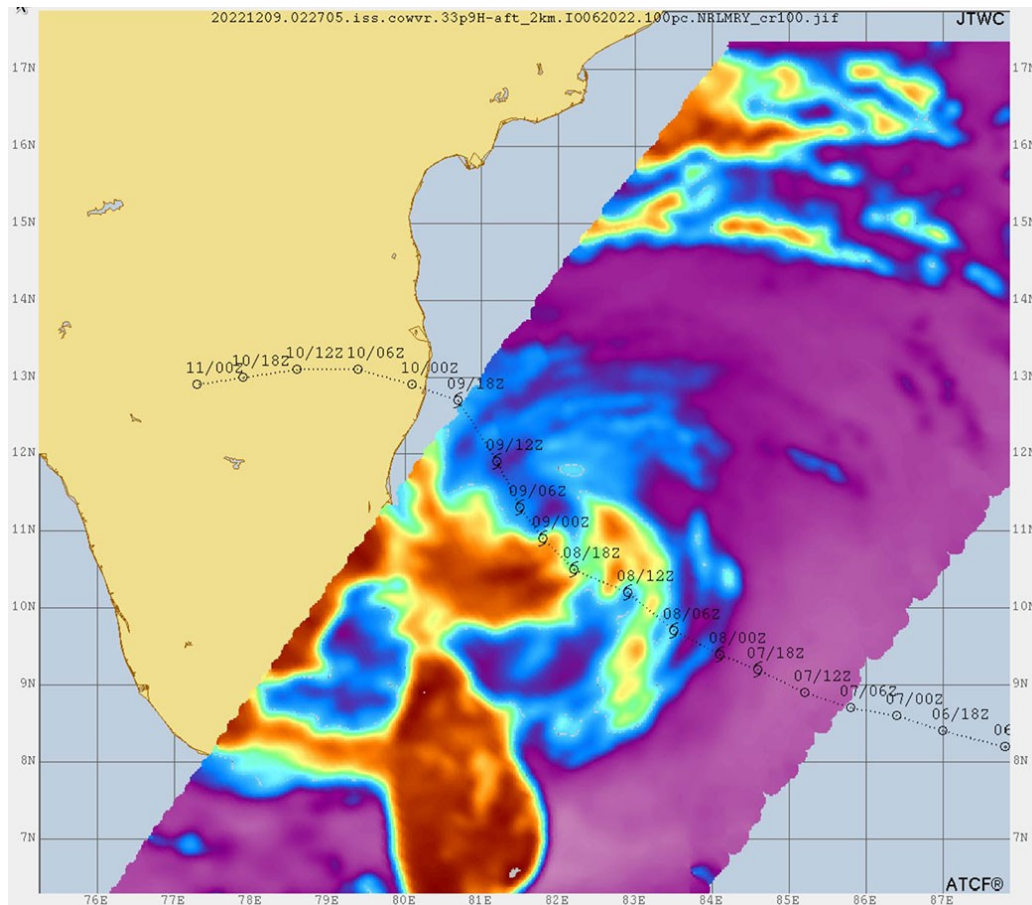
Credits: NASA

Known as COWVR and TEMPEST, the duo is demonstrating that smaller, less expensive science instruments can play an important role in weather forecasting.

NASA recently built two weather instruments to test the potential of small, low-cost sensors to do some of the work of bulkier, pricier satellites. Both instruments have exceeded expectations as trial runs, and they are already delivering useful forecast information for the most devastating of storms, tropical cyclones.

Launched in late 2021 to the International Space Station, COWVR (short for Compact Ocean Wind Vector Radiometer) measures the speed and direction of wind at the ocean surface, and TEMPEST (Temporal Experiment for Storms and Tropical Systems) provides atmospheric water vapor measurements. Both instruments are part of Space Test Program-Houston 8 (STP-H8), a three-year demonstration mission funded by the U.S. Space Force, which also funded the construction of COWVR. TEMPEST was built by NASA as a flight spare for a prior mission, and Space Force repurposed it for STP-H8.

Imagery created from their data is being used by the U.S. Joint Typhoon Warning Center to track the location and intensity of tropical cyclones in the Indian and Pacific oceans. In fact, COWVR and TEMPEST images were among the sources used by a forecaster at the typhoon center in Pearl Harbor, Hawaii, to pin down the location of Tropical Cyclone Mandous, which roiled the Bay of Bengal off southern India in December 2022.



Data from the COWVR and TEMPEST instruments aboard the International Space Station was used to create this image of Tropical Cyclone Mandous, which forecasters used to understand the December 2022 storm’s intensity and predict its path toward southern India.

Credits: U.S. Joint Typhoon Warning Center/U.S. Naval Research Laboratory

For several months, images based on COWVR and TEMPEST data have been delivered to the center by the Monterey, California-based Naval Research Laboratory (NRL), which has been working with NASA to calibrate the instruments and validate their data. Storm forecasters have been trying out the imagery – evaluating how it affects predictions and comparing it with other data sources – said Brian Strahl, the center’s director.

Reliable, frequently updated information on storm structure and location, wind speed, and humidity is crucial to the center’s mission to track tropical cyclones between Africa’s east coast and the west coast of the Americas, an area that includes vast expanses of open ocean.

“It’s challenging outside of the continental U.S. – where you don’t have weather aircraft routinely flying – to give a really good ground-truth of where these storms are, so we’re reliant on satellites,” Strahl said. “Any new additions of good quality data, which we believe these are, can be very useful.”

Smaller, Less Costly

COWVR and TEMPEST both measure microwave emissions from Earth’s atmosphere and surface. Data from microwave readings have an advantage over those from infrared or visible light: They can give forecasters a look at the internal structure of a tropical cyclone and help them locate the eye, even if it’s obscured by clouds.

The idea for the mission emerged a decade ago as the U.S. Department of Defense (DoD) started considering the next generation of instruments that could replace sensors such as WindSat, a DoD weather radiometer that operated until 2020.

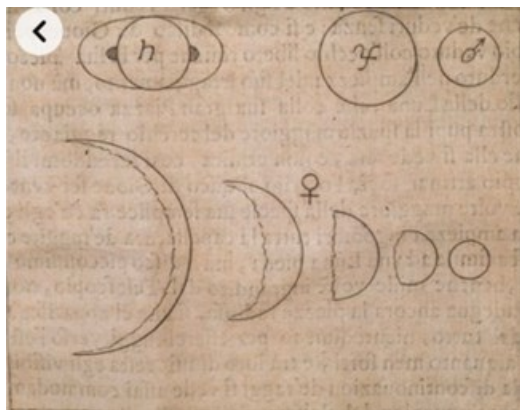


Kid Spot Jokes:

- ♦ **What is a light year?**
Same as a regular year but with less calories.
- ♦ **How did the astronaut serve drinks?**
In Sun glasses
- ♦ **How does the Solar system hold up its pants?**
With the Asteroid belt
- ♦ **When do astronauts have lunch?**
At launch time

Galileo's Telescopes, Observing Venus, Preserving Discoveries

Credit: Jordan Makower



Credit: solarsystem.nasa.gov



Credit: Personal

Galileo had at least 3 telescopes; the first one was only 8 power, equivalent to a standard binocular, but later improved to about 14, then 20 power. An important feature of the last one was that it had a long focal length (about 1330 mm.). Even though its aperture was only about one inch, the focal length produced an image of about $\frac{1}{4}$ degree, $\frac{1}{2}$ the apparent diameter of the Moon. With it, he was able to see detailed features on the lunar surface. He used it to view (among other things) Jupiter, Saturn, stars, and Venus. Those observations allowed him to formulate his theory that our planet and others revolved around the Sun. To reproduce his observations of Venus, great care must be used not to view the Sun in the telescope/binoculars used. Images could be projected onto a shaded flat surface for safety. Here's what he saw and recorded, in 1610.

Notice in the lower five images that Venus changes size and shape over a period of several months of observations, and it goes through phases, like our Moon. Galileo wanted to be recognized as the first person to observe this, and concealed his findings in a letter to a fellow astronomer using a coded Latin anagram: *Haec immatura a me iam frustra leguntur oy*, meaning "These unripe things are now read by me in vain." When deciphered, the phrase read: *Cynthiae figuras aemulatur mater amorum*, or "The mother of love imitates the forms of Cynthia." This meant that Venus goes through phases like the Moon. In 1631, Galileo observed Mercury; this was much more difficult since that planet is so close to the Sun. While Venus can be viewed up to 3 hours before sunrise/sunset, Mercury is at best seen 1-1.5 hours before sunrise or after sunset, and any object close to the sun should not be viewed through a simple telescope. USE GREAT CARE in projecting images with your optical device.

From the Board of Directors

Announcements

- ◇ Vini Carter, Ken Miura, Swami Nigam, Peter Melhus and Sukhada Palav were reelected into their respective Board of Director positions.
- ◇ Rob Jaworski was reelected as Treasurer. Aroon Nair was elected as President, Chris Gottbrath was elected as Vice-President and Swami Nigam was elected as Secretary. Their terms are effective until March 2024.
- ◇ Officer elections for President, Vice-President, Treasurer, and Secretary positions were held online via Electionbuddy and the new officers were announced at the board meeting on March 4. Please see article on Page 2 of this issue regarding the Officers.
- ◇ New QuickSTART Program Leader is Jonathan Lawton

Board Meeting Excerpts

December 10, 2022

In attendance

Wolf, Peter, Glenn, Rob
Absent: Sukhada, Ken, Vini
Guests: Emi, Marianne

SJAA minor's policy

The SJAA minor's policy was discussed with the board, and Peter and Wolf are to review and edit the minor's policy.

January 7, 2023

In attendance

Vini, Ken, Swami, Rob J, Peter, Glenn, Aroon, Sukhada
Absences: Wolf
Guests: Emi, Marianne, Larry, Jonathan, Kanch

End date and time for director or officer positions

Newly elected board members will take office after the February board meeting. Similarly, newly elected of-

ficers will take office after the March board meeting, and the SJAA bylaw document is to be updated. Vini moved to approve the action and Peter seconded. All approved.

Board Director 2

Ken, as President, appointed Aroon Nair as Director 2. Election for Director positions 1,3,5, 7, and 9 are planned for the February board meeting.

QuickStart Program Leader

Kanch introduced Jonathan Lawton as the new leader for the QuickSTART program. President Ken Miura appointed Jonathan as the program leader. For a headshot and a brief intro of Jonathan please look at Page 2 of this issue.

February 4, 2023

In attendance

Vini, Aroon, Ken, Wolf, Swami, Rob J, Peter, Sukhada
Absences: Glenn
Guests: Joe, Emi, Neeti, Nancy, Marianne, Kanch, John

Board member Elections

Results were announced for the Board of Directors Elections. 102 of 377 ballots were cast and there were 11 abstentions. There were 5 candidates for 5 vacancies - Ken Miura, Sukhada Palav, Vini Carter, Peter Melhus, and Swami Nigam. All incumbents were reelected with no new candidates. Their term is until February of 2025.

Astronomy Beginner's Forum

Wolf volunteered to be the leader of this new program which has the goal to educate people and introduce attendees to SJAA and its various programs but, at the same time, he said that he would welcome someone else to step up at a later time.

March 4, 2023

In attendance

Vini, Ken, Swami, Rob J, Peter, Glenn, Aroon, Sukhada, Wolf
Absences: None
Guests: Joe, Nancy, Marianne, Emi, Jonathan, Chris Gottbrath, Chis Wood, Nikola, Kanch

Officer Elections

Officer elections for President, Vice-President, Treasurer, and Secretary positions were held online via Electionbuddy and the new officers were announced at the board meeting on March 4. Aroon Nair was elected as President, Chris Gottbrath was elected as Vice-President, Swami Nigam was elected as Secretary and Rob Jaworski was re-elected as Treasurer. Their terms are effective until March 2024. For a headshot and a brief intro of Chris please look at Page 2 of this issue.

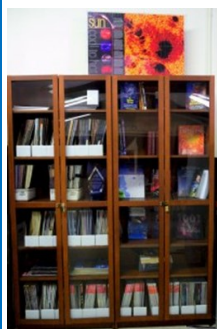
Funds approval for Solar program

Glenn moved to approve \$600 spending on Solar, Rob, Peter seconded. The motion was approved.

Printed newsletter for lifetime members

Marianne proposed that BOD approve Ephemeris and membership teams to approve printed newsletter for lifetime members at no cost, if they so desired. Peter moved to approve, Rob seconded, the board unanimously approved the motion.

SJAA Library



SJAA offers another wonderful resource: a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program, is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held monthly. The next Quick Start session during this time period is March 5th at Houge Park and March 18 at RCDO.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by Neeti Song, each with a similar format, with only the content changing to reflect what's currently in the night

sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group, this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Aroon Nair
Vice President:	Chris Gottbrath
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Swami Nigam
Director:	Vini Carter
Director:	Peter Melhus
Director:	Wolf Witt
Director:	Glenn Newell
Director:	Sukhada Palav
Director:	Ken Miura
Ephemeris Newsletter -	
Content Editor:	Abishek Ramasubramanian
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Hy Murveit
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Donations:	Rich Gregor
Quick START:	Jonathan Lawton
Solar:	Wolf Witt
Starry Nights:	Joseph Fragola
School Events:	Jon Anderson
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
Volunteer:	Nancy Lieu
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Place
postage
here

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **\$20** Regular Membership with online Ephemeris

☐ **\$30** Regular Membership with hardcopy Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name:

Address:

City/ST/Zip:

Phone:

E-mail address:
